

claims_evidence_matrix

Claims and Evidence Matrix

Authority state: CLAIM_HARDENED_DRAFT

Overall proof posture: partial_repo_verified

Reason: The uploaded packet contains mathematical exposition and architecture description, and the public repository was inspected at commit a756fa39e39342dc2443c6c7d9335f4c9d68455c. The repository verifies the operational app, defense/platform architecture, tick-based 3D simulation, and deployment scaffolding. Advanced physics, Riemannian geometry, mini-verse cosmology, and numerical validation claims remain evidence-gated because matching implementation files were not located in the inspected surface.

Claim Review Table

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-001	Series	Spatium Computationis models autonomous agents as physical entities in simulated spacetime.	C4, potential C1 for current 3D simulation substrate	Repo-backed for tick-based 3D entity simulation; advanced spacetime claims remain design/theory.	Medium	Cite simulation/world.py, simulation/engine.py, simulation/routes.py; add formal model for stronger claims.	“Spatium Computationis implements a tick-based 3D simulation substrate for agents, signals, threats, zones, and data packets; broader spacetime claims remain part of the research framing.”

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-002	Paper 1	The physics engine implements Velocity Verlet integration.	Potential C1	Not located in inspected repo surface.	High	Specific physics-engine file, tests, and run logs.	"The canonical paper proposes a Velocity Verlet physics layer; the inspected repo currently verifies a tick-based 3D simulation layer."
C-003	Paper 1	Verlet integration yields bounded energy error for Hamiltonian systems.	C10, C9	Externally supportable with citations.	Low	Cite Verlet and geometric numerical integration sources.	"Under standard assumptions, symplectic methods such as Velocity Verlet are used for bounded long-term energy behavior."
C-004	Paper 1	The system enforces a hard speed bound at c .	C3, potential C1	Not located in inspected repo surface.	High	Relativistic update tests and edge-case traces.	"The research design includes a relativistic correction intended to bound simulated propagation speed."
C-005	Paper 1	Conservation error remains below 10^{-6} for typical time steps.	Potential C1	Not located in inspected repo surface.	High	Run logs, plots, parameter definitions.	"Validation should report conservation error under specified time-step and parameter regimes."

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-006	Paper 2	Geometry quantities are computed numerically from metric functions.	Potential C1, C9	Not located in inspected repo surface.	High	Code and benchmark metrics.	“The geometry design uses metric functions as the interface for numerical differential geometry.”
C-007	Paper 2	The engine works for any smooth metric.	C9	Overbroad without numerical constraints.	High	Stability domain, convergence tests, singularity handling.	“The approach targets smooth metric functions within defined numerical stability limits.”
C-008	Paper 2	Horizon detection can be performed by scanning sign changes in g_{tt} .	C10, potential C1	Conceptually supported but coordinate-sensitive.	Medium	Benchmark metrics and caveat discussion.	“For suitable coordinates, horizon candidates may be detected by monitoring metric component behavior such as sign changes in g_{tt} .”
C-009	Paper 3	Mini-verses can use configurable constants and topologies.	C8, potential C1	Not located in inspected repo surface.	High	Schema, examples, tests.	“The research architecture defines mini-verse instances with configurable constants and topology choices.”

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-010	Paper 3	Entropy tracking provides a thermodynamic arrow of time.	C3, C9	Needs formalization.	Medium	Entropy model and numeric examples.	"Entropy accounting is proposed as an indicator of thermodynamic directionality."
C-011	Paper 3	Inter-verse channels enforce causality through bandwidth limits.	C8, C3	Not located in inspected repo surface.	High	Transfer model, tests, causal proof sketch.	"The architecture specifies bandwidth-limited inter-verse channels intended to constrain transfer."
C-012	Paper 4	Agents interact through physical forces where attraction models influence and electromagnetic forces model communication.	C4, C9	Metaphorical/prototypical claim.	Medium	Formal mapping from physical variables to operational semantics.	"The architecture uses physical analogies to represent influence and communication relationships."
C-013	Paper 4	The battleground architecture contains cooperative, hostile, and shadow routing tiers.	C5, C8, partial C1 for documented routes	Repo-backed architecture and route surfaces; release-gated for sensitive details.	Medium	Cite README.md, main.py, defense/honeypot/router.py, platform_router.py.	"A tiered battleground architecture is documented and partially implemented through defense, honeypot, platform, and simulation routes."

ID	Paper	Claim summary	Claim class	Current posture	Risk	Needed support	Recommended wording
C-014	Series	The system is production-grade.	Potential C1	Deployment scaffolding is repo-backed; production operation still needs live deployment/CI run evidence.	Medium	Workflow run status, release artifact, deployment URL, operational logs.	“The repository contains production-oriented deployment scaffolding, including Docker, compose, and CI build/test workflow; live production status requires runtime evidence.”

Citation and Evidence Gaps

Missing external citations:

- Symplectic integration and long-term energy behavior.
- Numerical differential geometry and finite-difference curvature computation.
- General relativity references for benchmark metrics and invariants.
- Cosmological evolution and entropy references.
- Agent-based modeling, cyber range, sandbox, and immune-system-inspired defense literature.

Repository evidence status:

- Repository identified by operator: <https://github.com/FreddyCreates/pegasus-battleops>.
- Runtime access status: GitHub connector succeeded; shell clone remained blocked by the container network.
- Commit inspected: a756fa39e39342dc2443c6c7d9335f4c9d68455c.
- Promotion rule: implementation claims may be upgraded only when tied to specific files, commits, tests, and logs.

Missing internal evidence still needing inspection:

- File-level implementation evidence for advanced physics, geometry, and mini-verse claims.
- Test suite contents and actual output logs.
- Benchmark results.
- Diagrams or screenshots of the runtime/visualization if visual claims are retained.

Missing experiments or validations:

- Energy/momentum/angular momentum conservation traces.
- Geodesic norm preservation and analytic comparison tests.

- Horizon detection benchmark.
- Mini-verse topology behavior examples.
- Agent routing and containment simulation traces.

Claims to remove or downgrade before public release:

- “production-grade” should be narrowed to “production-oriented deployment scaffolding” unless live deployment, passing CI run, or operational evidence is attached.
- “all implementations use real SI units” unless unit tests or code evidence are attached.
- “validated conservation laws” unless actual validation output is included.
- “proper differential geometric constructions” should be reframed as “standard mathematical constructions are specified” unless implementation tests are attached.

Next Proof Move

Use the repo-backed architecture and simulation evidence to strengthen Paper 4 immediately, and either locate the missing advanced engine files or reframe Papers 1-3 as staged theory/design papers aligned with the current tick-based 3D simulation substrate.